

Group 5.4 Social Problem Report

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2026-06-17

Title Page

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Part 1 - Identifying a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

Education and Income Inequality in the Netherlands Income inequality is an important societal concern in the Netherlands. Differences in income influence various aspects of life, such as level of living, healthcare, education, therefore research demonstrated how individual's place of residence and educational background can predetermine socioeconomic prospects for the future. As a result, people with similar abilities may experience different economic opportunities depending on their educational background and geographic location.

Key factor associated with income is level of education. Higher education may lead to higher prospects for employment and better wages. Regional exposure may also influence income through variations in labor market conditions, economic development, and access to educational resources. If income depends strongly on education level and region, policymakers can develop targeted measures to reduce inequalities and promote equal opportunities across different parts of the country.

In this report we quantify the relationship between level of education, region and income in the Netherlands using data from Statistics Netherlands (CBS). Specifically, we analyze how income varies across different education levels and regions and assess whether these factors are associated with economic outcomes.

Part 2 - Data Sourcing

2.1 Load in the data

Dataset source:

- CBS Dataset 86162NED – Income statistics
- CBS Dataset 85525NED – Educational attainment by region

2.2 Short summary of the dataset(s)

Education dataset: 70585 observations, 21 variables

Dataset 1: Educational Attainment (85525NED)

The dataset “Bevolking; hoogstbehaald onderwijsniveau en regio” contains information on the highest educational attainment of people aged 15–75 in the Netherlands, broken down by region, gender, age group, and year. The data are collected by Statistics Netherlands (CBS) and are available from 2013–2024. The unit of observation is a combination of region and year.

Key variables include: - the percentage of people with low education (primary education, VMBO, MBO1), - medium education (HAVO, VWO, MBO2-4), and higher education (HBO, WO).

For this project, the share of highly educated people is the most relevant variable, as it can be compared with regional income levels. A limitation of the dataset is that it contains aggregated regional statistics rather than individual-level data, so relationships observed across regions may not necessarily apply to individuals.

Income dataset: 419244 observations, 11 variables

Dataset 2: Personal Income (86162NED)

The dataset “Inkomen van personen; persoonskenmerken, regio” (CBS) contains information on personal income by region and demographic characteristics in the Netherlands. The unit of observation is a combination of region, year, gender, and personal characteristics.

Key variables include: - RegioS: region code - Perioden: year - GemiddeldPersoonlijkInkomen: average personal income (1,000 euros) - MediaanPersoonlijkInkomen: median personal income (1,000 euros)

The dataset covers the period 2011–2024. Figures for 2024 are provisional. A limitation of the dataset is that it contains aggregated regional data rather than individual-level observations. For our analysis, the most relevant variable is median personal income or average personal income, because these can be compared with regional educational attainment. A limitation is that the dataset is aggregated by region, so it does not allow conclusions about individual people. Also, the 2024 figures are provisional, and regional averages may hide differences within municipalities or provinces.

These datasets can be merged using the region (RegioS) and year (Perioden) variables, allowing us to examine the relationship between educational attainment and personal income across Dutch regions.

Dataset	Unit of observation	Key variables	Source
Onderwijs (85525NED)	One Dutch region in one year	Region, year, population size, share low education, share medium education, share high education	Statistics Netherlands (CBS StatLine), 2013–2024
Inkomen (86162NED)	One Dutch region in one year	Region, year, average personal income, median personal income, number of persons with income	Statistics Netherlands (CBS StatLine), 2011–2024

2.3 Describe the type of variables included

Educational Attainment Dataset (data.onderwijs) (85525NED)

This dataset primarily contains socio-economic status (SES) information, as educational attainment is a common indicator of an individual's socio-economic position. The dataset includes variables on the highest completed level of education, population counts, age groups, gender, region, and year.

The data are obtained from administrative sources and registers maintained by Statistics Netherlands (CBS). Educational attainment is compiled using information from educational institutions and government records. The dataset is aggregated at the regional level, meaning that it reports percentages and population counts for groups rather than information on individual persons.

Income of Persons Dataset (data.inkomen) (86162NED)

The variables in this dataset mainly contain socio-economic status (SES) information, as they measure personal income levels across different regions and population groups. Key variables include average personal income, median personal income, the number of persons with income, region, year, gender, and other demographic characteristics.

The data are derived from administrative sources, such as tax records and social security registers, and are compiled by Statistics Netherlands (CBS). The information is therefore not collected through surveys or interviews. The dataset is aggregated by region and demographic group, meaning that it provides summary statistics rather than individual-level income data.

Part 3 - Quantifying

3.1 Data cleaning

```
## Income missing values: 104576

## Income duplicates: 0

## Education missing values: 225363

## Education duplicates: 7

## Income dataset:

## 419244 observations and 11 variables

## Education dataset:

## 70585 observations and 21 variables

## Income observations after cleaning: 386830

## Education observations after cleaning: 57082
```

Filtering for data.inkomen

RegioS	Perioden	GemiddeldPersoonlijkInkomen_6
LD01	2011JJ00	9.7
LD01	2012JJ00	9.5
LD01	2013JJ00	9.3

Cleaning steps for the income dataset:

Trimmed whitespace from Geslacht, Persoonskenmerken, RegioS, and Perioden columns, since CBS exports sometimes include trailing spaces that prevent exact matching during the merge. Filtered to Geslacht == “T001038” (total population) to avoid double-counting separate male/female rows. Filtered to Persoonskenmerken == “53050” to select the relevant personal characteristics category matching the education dataset’s age grouping, keeping the two datasets comparable (25-45 age). Filtered RegioS to keep only LD (COROP region) and PV (province) codes, since the analysis focuses on provincial-level comparisons and municipality-level rows would not match the education dataset’s geography. Selected only the three columns needed for analysis (RegioS, Perioden, GemiddeldPersoonlijkInkomen_6)

Chosen variables:

RegioS — region identifier Perioden — time period GemiddeldPersoonlijkInkomen_6 — mean personal income ($\times 1,000$ euros)

Filtering for data.onderwijs

RegioS	Perioden	k_2HavoVwoMbo24_3	k_3HboWo_4	k_32HboWoMasterDoctor_9
AM01	2016JJ00	48.90	23.3	7.3
AM01	2017JJ00	47.45	24.4	8.1
AM01	2018JJ00	47.20	25.2	8.3

Cleaning steps for the education dataset:

Filtered to Geslacht == “T001038” (total population, both genders combined) to avoid double-counting gender subgroups. Filtered Leeftijd to age groups 53050 (25-35 age) and 53700 (35-45 age) (the two adult age brackets relevant to labour market participation) and averaged them, since keeping both separately would create duplicate region-year rows. Trimmed whitespace from RegioS and Perioden, since CBS region codes sometimes contain trailing spaces that would otherwise prevent a correct join with the income dataset. Aggregated to one row per region-year using group_by() and summarise(), averaging the three education share variables.

Chosen variables:

RegioS — region identifier Perioden — time period k_2HavoVwoMbo24_3 — medium education share (%) k_3HboWo_4 — higher education share, bachelor level (%) k_32HboWoMasterDoctor_9 — higher education share, master/doctor level (%)

Merging two data sets

```
## [1] 192
```

```
## [1] 6
```

##	RegioS	Perioden	GemiddeldPersoonlijkInkomen_6	k_2HavoVwoMbo24_3	k_3HboWo_4	k_32HboWoMasterDoctor_9
## 1	LD01	2013JJ00	9.3	48.70	18.75	5.00
## 2	LD01	2014JJ00	9.2	48.35	19.40	5.15
## 3	LD01	2015JJ00	9.3	48.20	19.85	5.25

## 4	LD01 2016JJ00	9.5	48.20	19.80	5.40
## 5	LD01 2017JJ00	9.8	47.35	20.55	5.70
## 6	LD01 2018JJ00	10.3	47.25	21.40	5.95

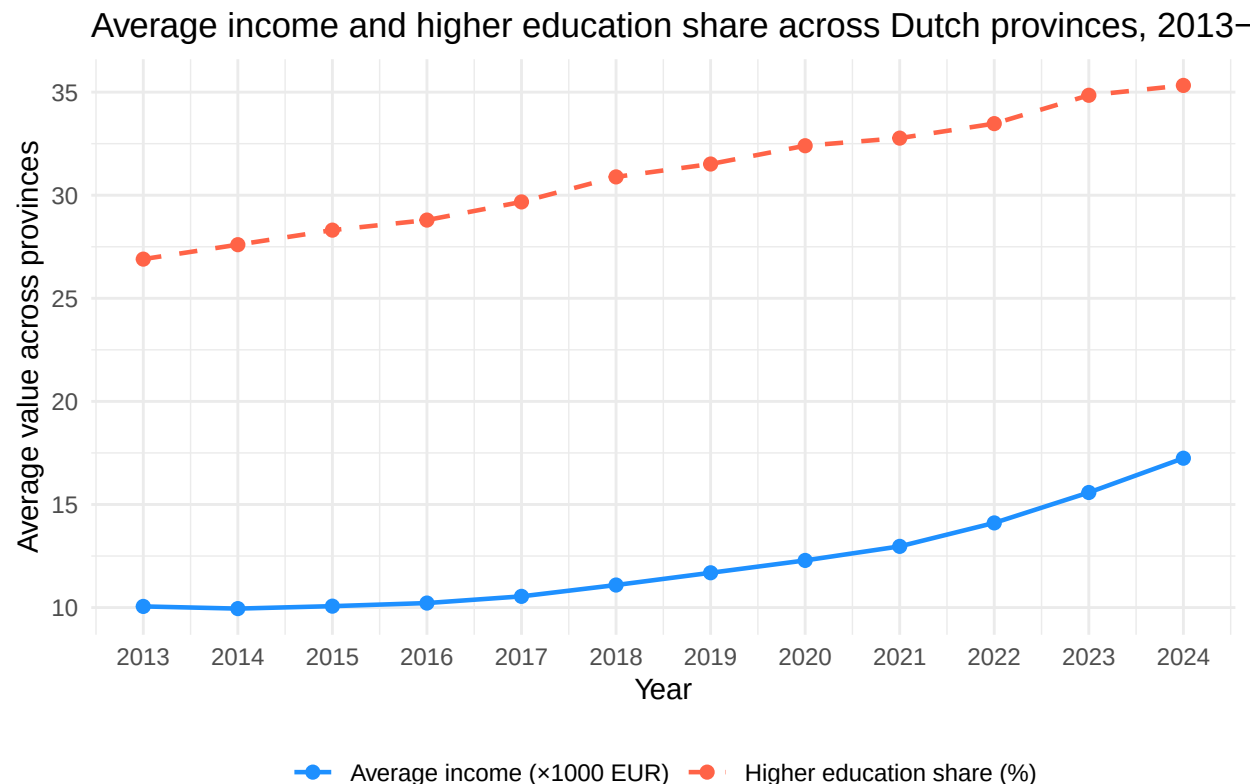
The two cleaned datasets are merged using an inner join on RegioS and Perioden, which together uniquely identify a region-year observation in both datasets. An inner join is used rather than a left or full join because only region-year combinations present in both the income and education datasets are meaningful for this analysis — rows that exist in only one dataset would introduce missing values and cannot be used to study the relationship between education and income.

The merged dataset (data.combined) contains 192 observations and 6 variables, one row per region per year, which serves as the basis for all subsequent analysis.

3.2 Generate necessary variables

1. **high_education_share** — sum of bachelor and master/doctor shares, giving the total percentage of the population with higher education in a region. Used in the temporal, spatial, and sub-population charts.
2. **mid_education_share** — share of the population with HAVO/VWO/MBO-level education, used as a comparison group in the sub-population chart to show how mid-level education differs from higher education across provinces.
3. **income_growth** — year-on-year percentage change in average income, calculated as $(\text{current year income} - \text{previous year income}) / \text{previous year income} \times 100$. Used in the event analysis to detect whether income growth accelerated or slowed around COVID-19.

3.3 Temporal variation — average income and education share over time



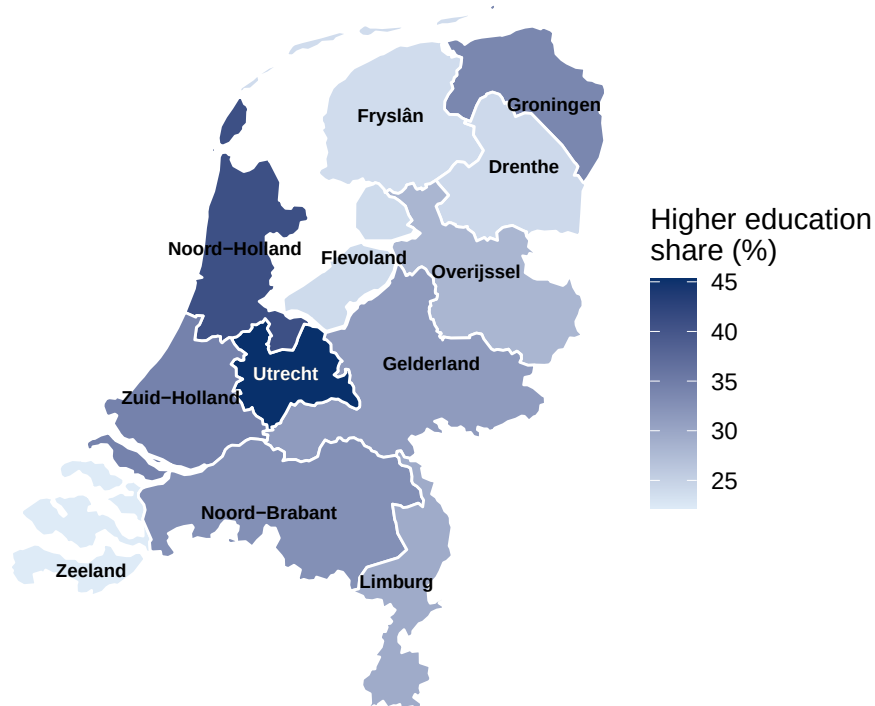
Source: Statistics Netherlands (CBS).

Between 2013 and 2024, both average personal income and the share of higher-educated residents increased steadily across Dutch provinces. Average income rose from approximately €10,000 to €17,000, while the higher education share grew from 28% to 37%. The two trends move in parallel, suggesting a positive association between education levels and income over time. Growth in both indicators accelerated after 2018. The dotted line marks the onset of COVID-19 in 2020; while no sharp break is visible, income growth slowed slightly in 2019–2020 before recovering strongly. Note that rising income partly reflects inflation rather than real income gains, and the parallel trends do not establish a causal relationship — both may reflect broader urbanisation and economic growth.

The province boundary shapefile is joined with the averaged income and education data using `statnaam` (province name) as the join key, after first matching CBS region codes to province names via `prov_lookup`.

3.4 Visualize spatial variation

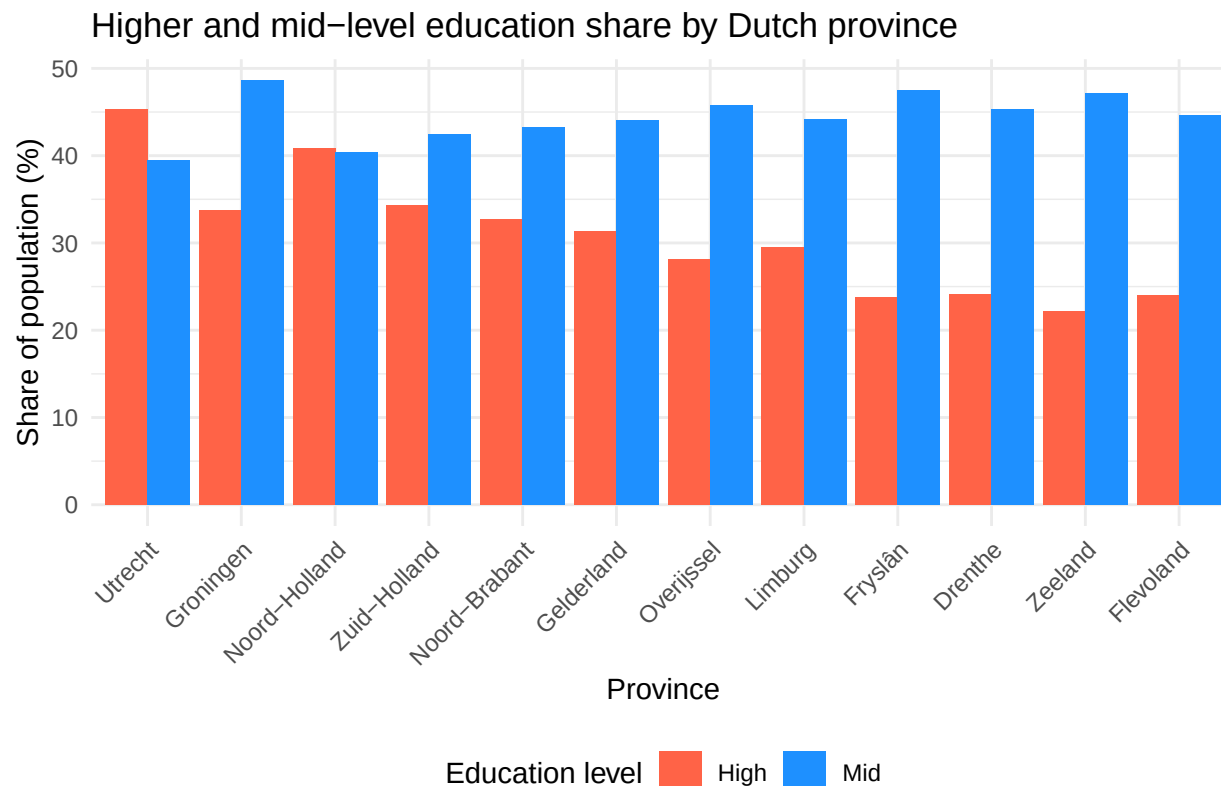
Average higher education share by Dutch province, 2013–2024



Source: Statistics Netherlands (CBS). Boundary: CBS/cartomap 2024.

The map reveals a clear urban-rural divide in higher education across Dutch provinces. Utrecht (~45%) and Noord-Holland (~41%) stand out as the most highly educated, reflecting the concentration of universities and knowledge-intensive industries in the Randstad. In contrast, peripheral provinces like Zeeland (~22%), Fryslân (~24%) and Drenthe (~24%) have the lowest shares. Groningen is a notable exception among northern provinces, likely due to its large university. Provinces with higher education shares generally also have higher average incomes, suggesting education partly drives regional income differences. However, Zeeland — the least educated province — records the highest average income (~€14k), indicating that education alone does not fully explain provincial income variation.

3.5 Visualize sub-population variation

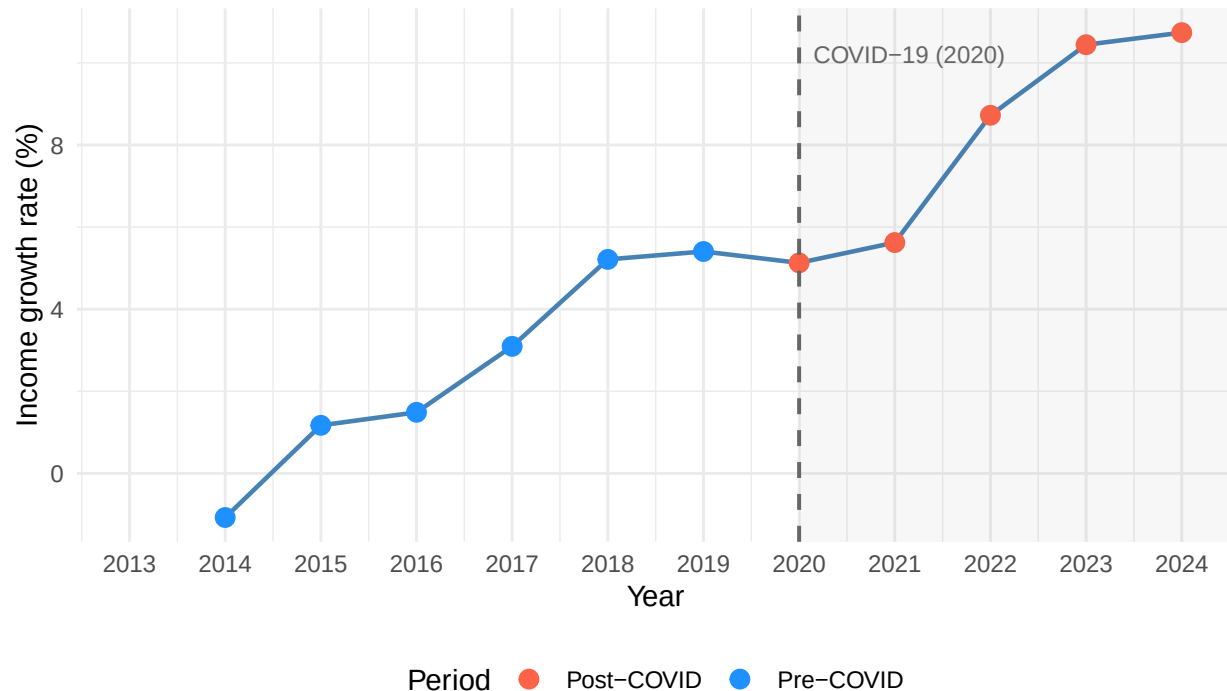


Source: Statistics Netherlands (CBS).

This grouped bar chart is directly relevant to understanding regional income inequality in the Netherlands. If higher education is a key driver of personal income — as suggested by the scatter plots in sections 3.5 and 3.6 — then provinces with low higher education shares are structurally disadvantaged in terms of earning potential. The large variation in higher education share across provinces (22% in Zeeland to 45% in Utrecht) implies that where someone is born or lives in the Netherlands significantly affects their access to high-income opportunities. Mid-level education, by contrast, is evenly distributed across provinces and therefore cannot explain regional income differences. This means that policies targeting higher education access in peripheral provinces like Zeeland, Fryslân and Drenthe could be an effective lever for reducing the income gap between Dutch regions.

3.6 Event analysis

Annual growth rate of average personal income before and after COVID-19
Dutch provinces, 2013–2024



Source: Statistics Netherlands (CBS).

Prior to COVID-19, income growth was modest and gradually increasing, averaging around 1–5% annually. Contrary to expectations, the onset of COVID-19 in 2020 did not reduce income growth — the Dutch government’s NOW-regeling wage subsidy scheme cushioned the impact on household incomes. More striking is the sharp post-COVID acceleration, with growth rates nearly doubling from ~5% in 2020 to ~10% by 2024, likely driven by inflation compensation and a tight labour market particularly benefiting higher-educated workers. Also to note that this is an association, not a causal claim — the upward trend in income growth began before 2020, suggesting broader structural factors beyond COVID are at play.

Part 4 - Discussion

4.1 Discuss your findings

Combining the four visualisations provides a coherent picture of income and education in the Netherlands:

First, both income levels and educational attainment increased over time. Second, strong geographical differences exist, with provinces in and around the Randstad generally displaying higher education levels than more peripheral regions. Third, differences between educational sub-populations indicate that variation in higher education attainment is much larger than variation in mid-level education. Finally, while income growth accelerated after the COVID-19 period, the data do not allow causal conclusions regarding the effect of the pandemic itself. Taken together, the results suggest that educational attainment is positively associated with income outcomes and that regional differences in education may contribute to broader patterns of economic inequality within the Netherlands. But because the analysis uses aggregated regional data, the observed associations cannot be interpreted as individual-level causal effects.

Part 5 - Reproducibility

5.1 Github repository link

Part 5 – Reproducibility

GitHub repository link

The complete project, including all code, documentation, and generated figures, is available at:

<https://github.com/ernestveyenberg/ProgrammingForEcon---Group-5-4>

How this project is reproduced

To reproduce the analysis:

1. Clone the GitHub repository.
2. Open the R project.
3. Restore packages using `renv::restore()`.
4. Run the data preparation scripts.
5. Knit report.Rmd to generate the final PDF report.

5.2 Reference list

Use APA referencing throughout your document.